

Business Problem Statement

A retail organization aims to gain deeper insights into customer purchasing behavior to improve business performance, enhance customer satisfaction, and strengthen customer retention. Recent shifts in buying patterns across different age groups, product categories, locations, and shopping preferences have created a need for data-driven decision-making. The company seeks to understand the key factors that influence customer purchases, including product reviews, discounts, seasonal trends, payment methods, and purchase frequency.

The primary objective of this project is to analyze customer shopping data and generate actionable insights that can support marketing, sales, and customer engagement strategies.

Key Business Question

How can customer shopping data be utilized to identify purchasing trends, improve customer retention, and optimize product offerings and marketing campaigns to drive business growth?

Project Deliverables

1. Data Cleaning & Transformation (Python)

- Import and preprocess the raw customer dataset.
- Handle missing values, inconsistencies, and data quality issues.
- Create meaningful features and transform data for analysis.
- Prepare a clean and analysis-ready dataset.

2. Database Design & Analysis (SQL)

- Store the processed data in a relational database.
- Write SQL queries to explore customer behavior and purchasing patterns.
- Analyze customer demographics, spending habits, and loyalty indicators.
- Generate business-focused insights through structured data analysis.

3. Dashboard Development (Power BI)

- Build an interactive and user-friendly dashboard.
- Visualize customer trends, sales performance, and key metrics.
- Enable stakeholders to explore insights through dynamic reports and filters.
- Support data-driven business decisions with clear visual storytelling.

4. Business Insights & Recommendations

- Interpret analytical findings and identify growth opportunities.
- Provide recommendations for customer retention and engagement.
- Suggest strategies for improving sales, promotions, and product performance.
- Summarize key outcomes in a professional project report.

5. GitHub Repository Documentation

- Maintain a well-organized repository containing Python scripts, SQL queries, Power BI dashboard files, and project documentation.
- Ensure reproducibility and ease of collaboration for future development.